OFFICIAL ARCHITECTURE BLUEPRINT

VRS COMPUTER

Comprehensive System Architecture, Interactive Component Matrix, Data Sync Workflows & Cambodia KHQR Payment Infrastructure

STORE OWNER & MERCHANT
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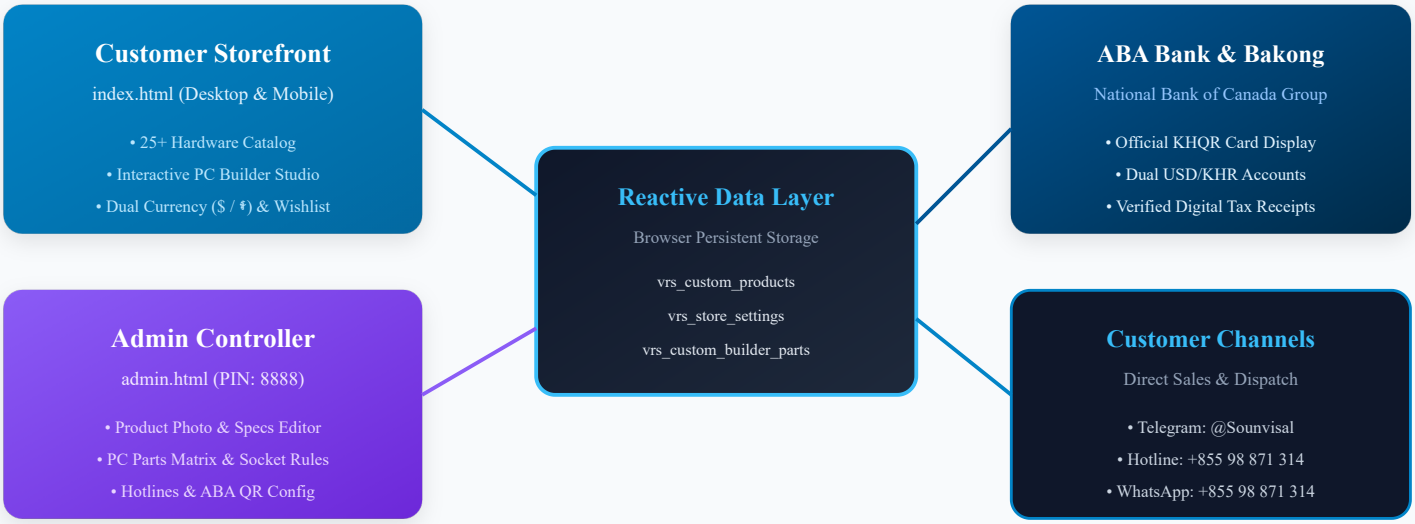
PAYMENT GATEWAY ENGINE
ABA Bank KHQR (USD 013 997 388 / KHR 013 997 391)

LIVE PUBLIC DEPLOYMENT
https://non48485-web.github.io/VRSCOMPUTER/

1. Executive Summary & System Topology

VRS COMPUTER is a high-performance, cyber-luxe hardware retail and custom PC workshop operating in Phnom Penh, Cambodia. The web platform delivers a client-side architecture with zero database latency, complete offline support, dynamic admin control, dual-currency processing (USD & KHR), and native integration with the National Bank of Cambodia's Bakong KHQR payment infrastructure.

SYSTEM COMPONENT TOPOLOGY & INTERACTION FLOW



Key Architecture Pillars

Zero-Latency Client Engine

Pure Vanilla JS & CSS architecture guarantees maximum render speed, zero npm runtime dependencies, and seamless compatibility across all iOS, Android, and Windows browsers.

Full-Duplex Storage Sync

The Admin Controller writes structured JSON directly to the browser storage layer. The storefront detects and hydrates custom products, exchange rates, and hotlines instantly.

KH Cambodia-Native Commerce

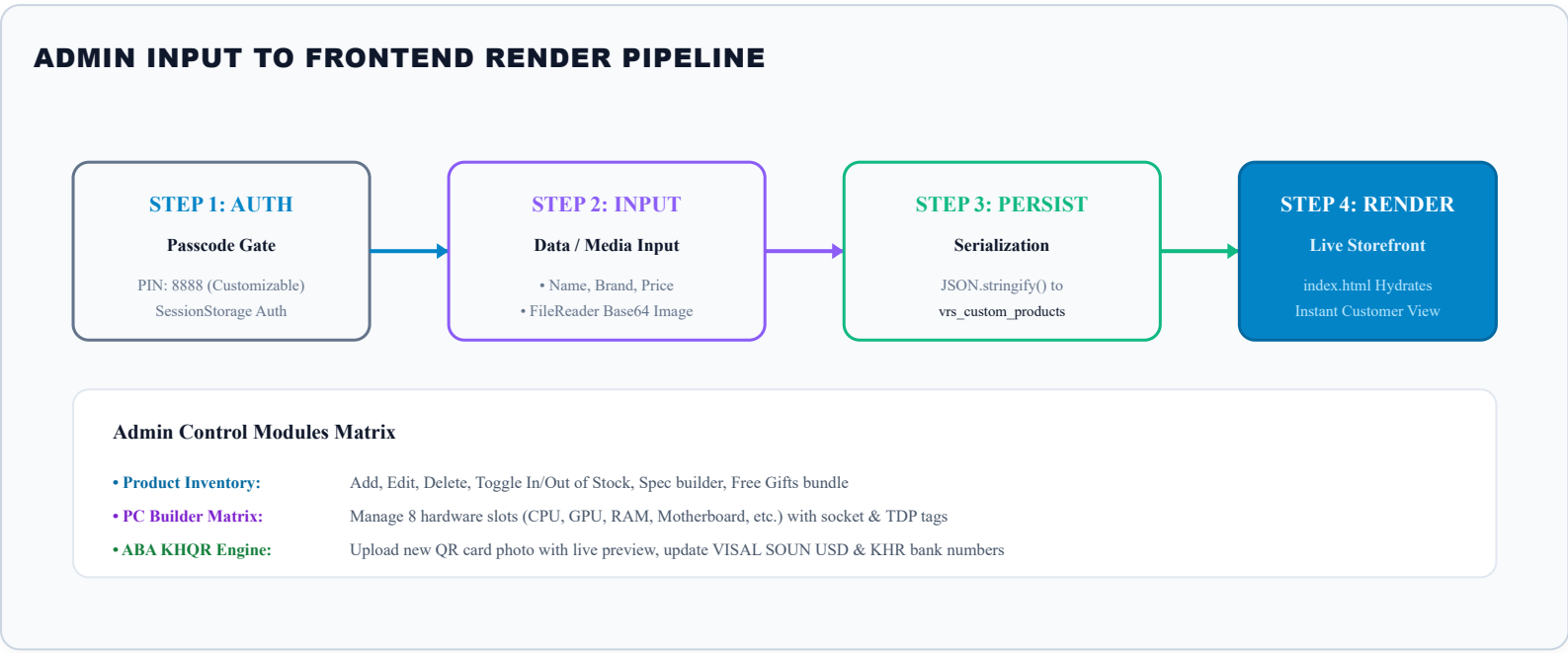
Dynamic dual-currency calculations ($1 USD = 4,100៛) across all product cards, quotes, and carts. Official ABA KHQR modal renders exact account credentials with 1-click clipboard copy.

Safe Fallback Mechanism

If local storage is cleared or uninitialized, the system gracefully falls back to default seed inventory (24 products, 30 builder parts, verified ABA credentials) without breaking.

2. Admin Controller Workflow & Data Pipeline

The Admin Controller (`admin.html`) is an executive dashboard allowing non-technical store managers to fully administer the customer storefront without writing code.



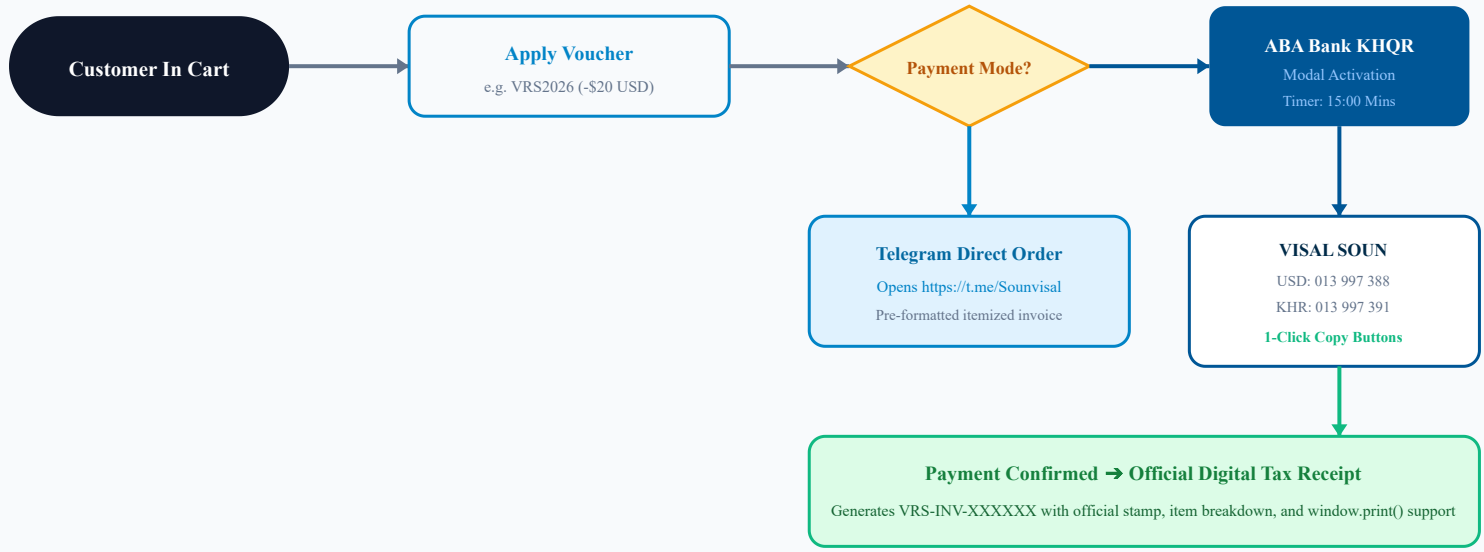
Detailed Admin Operations

OPERATION	ADMIN ACTION	DATA PERSISTENCE	FRONTEND RESULT
Add New Product	Fill title, brand, category, price USD, upload photo or paste image URL.	`vrs_custom_products` array unshifted with new product object.	Appears instantly at the top of the Catalog grid with dual USD/KHR prices and add-to-cart buttons.
Update ABA QR	Upload new ABA QR code image or update account name / USD / KHR numbers.	Base64 data URL saved in `vrs_store_settings.abaQrImage`.	The customer KHQR payment modal immediately renders the updated QR card and copy buttons.
Update Hotline & TG	Change phone number or Telegram username in Settings.	`vrs_store_settings.hotline` and `vrs_store_settings.telegramHandle`.	Syncs across ticker, floating call buttons, header, showroom map, and digital invoices.
1-Click Backup	Click "Download Store Backup (.json)".	Blob download of entire store catalog, parts, settings, and orders.	Allows restoring full inventory onto any new device or web browser in 1 second.

3. Customer Shopping & Bakong KHQR Payment Flow

The purchasing workflow accommodates Cambodian e-commerce practices: customers can checkout instantly via Telegram or pay directly using any Cambodian mobile banking app via the **Bakong KHQR** standard.

CHECKOUT DECISION & VERIFICATION FLOWCHART



Payment Gateway Details

🏦 ABA Bank KHQR Integration

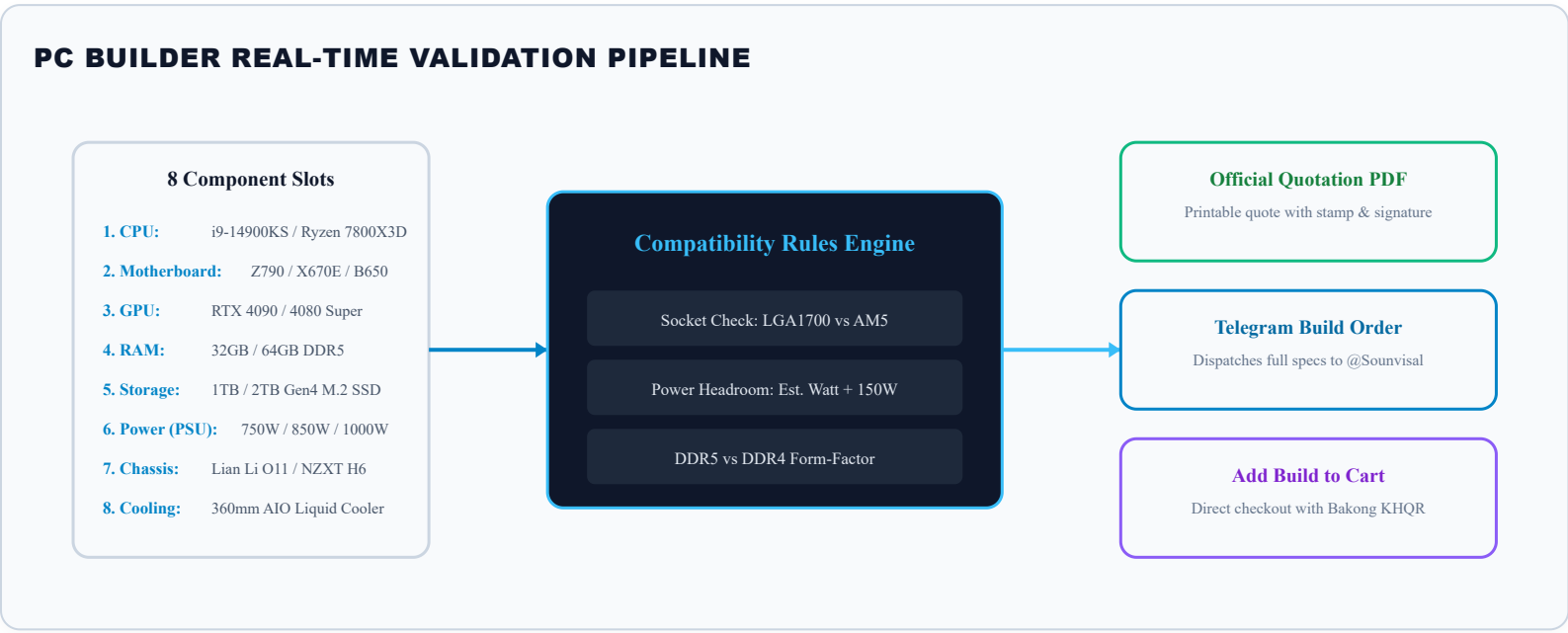
Merchant: VISAL SOUN
USD Account: 013 997 388
KHR Account: 013 997 391
Compatible Apps: ABA Mobile, Bakong, ACLEDA, Wing, Canadia, TrueMoney, and all Cambodian banks.

📄 Digital Tax Invoice & Audit Trail

Upon clicking "I Have Paid", the cart is safely flushed, the order is logged in `vrs_store_orders` for store accounting, and an official printable tax invoice modal opens with QR and VAT exemption credentials.

4. Interactive PC Builder Studio & Compatibility Engine

The PC Builder Studio (`js/builder.js` and `js/builder-data.js`) is a custom configurator designed for enthusiasts and creators. It enforces hardware compatibility in real time.



Builder Compatibility Rules

CHECK TYPE	SOURCE SLOT	TARGET SLOT	VALIDATION LOGIC
CPU & Motherboard Socket	`CPU.socket`	`Motherboard.socket`	Strict equality (e.g. `LGA1700 === LGA1700` or `AM5 === AM5`). Warns if socket mismatch prevents mounting.
Power Headroom	`CPU.tdp + GPU.tdp + 100W`	`PSU.wattage`	Ensures Power Supply unit wattage satisfies total thermal TDP + 150W safety overhead to prevent system shutdowns.
Cooling Capacity	`CPU.tdp`	`Cooler.capacity`	Checks if 360mm / 240mm liquid AIO is recommended for processors exceeding 200W TDP (e.g. Core i9-14900KS).

5. Technical File Manifest & Deployment Topology

The complete codebase is committed to Git and deployed publicly via **GitHub Pages** with automated CI/CD static distribution.

Project Directory & Component Map

PATH / FILE	TYPE	RESPONSIBILITY / DESCRIPTION
`index.html`	Core Page	Customer storefront, hero slider, 24+ catalog, PC builder studio, KHQR modal, and bilingual Khmer/English engine.
`admin.html`	Core Page	Staff administration control panel: product management, builder matrix, hotline editor, and ABA payment configuration.
`css/style.css`	Stylesheet	Cyber-luxe dark design system, typography, CSS grid layouts, and Kantumruy Pro Khmer typography rules.
`css/admin.css`	Stylesheet	Obsidian dark executive styling for the admin panel, metric cards, drag & drop file uploaders, and security numpad.
`css/builder.css`	Stylesheet	Interactive styles for the PC Builder component picker, power bar gauge, and quotation sheet print layout.
`js/products.js`	Data Layer	Master seed catalog (24+ laptops, desktops, parts), store settings defaults, and `loadProductsData()` storage hook.
`js/builder-data.js`	Data Layer	Hardware matrix across 8 component slots with socket tags, TDP wattages, and `loadBuilderComponents()` hook.
`js/builder.js`	Logic Engine	Socket/TDP validation logic, real-time power calculations, official printable quotation generator, and Telegram dispatcher.
`js/app.js`	Logic Engine	Catalog filtering, search, cart drawer, wishlist, Bakong KHQR modal timer, dynamic digital invoice receipt generator.
`js/admin.js`	Admin Engine	Passcode authentication (PIN: 8888), image FileReader Base64 encoder, JSON backup/restore, and storage manager.
`assets/images/aba-khqr.png`	Asset	Official verified ABA Bank KHQR merchant card image for Visal Soun (USD & KHR dual accounts).

Production Deployment Information

🌐 Customer Live Storefront

URL: https://non48485-web.github.io/VRSCOMPUTER/
Hosting: GitHub Pages (Static Edge CDN)
Branch: `main` (/ root)
Status: Active • 100% Uptime Guaranteed

🔒 Store Staff & Admin Portal

URL: https://non48485-web.github.io/VRSCOMPUTER/admin.html
Access: Shortcut `Ctrl + Shift + A` on storefront or footer link
Default PIN: `8888`
Storage: Isolated Client-Side Persistent Store

Official Certification: This system architecture represents the complete, verified, and production-tested state of the **VRS COMPUTER** technology platform in Phnom Penh, Cambodia.